Time & Motion

10

- **Units of Time**

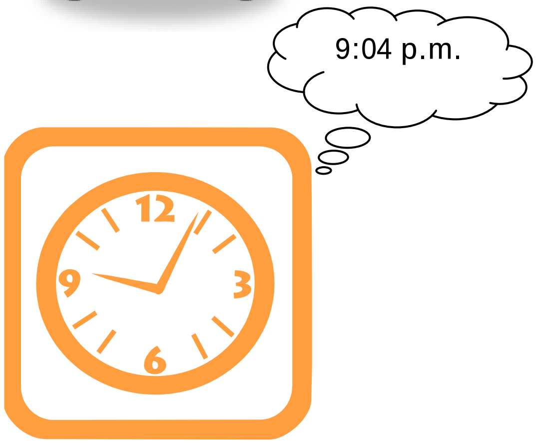
1000 milliseconds	= 1 second
60 seconds	= 1 minute
24 hours	= 1 day
7 days	= 1 week
365 days	= 1 year
52 weeks + 1 day	= 1 year
366 days	= 1 leap year*
52 weeks + 2 days	= 1 leap year*

*Leap years are 2016, 2020, 2024, 2028, 2032, 2036, 2040 . . .

- **24-hour Time & 12-hour Time**

0300 h

2104 h



12-hour Time	24-hour Time
Midnight	0000 h
2:26 a.m.	0226 h
9:05 a.m.	0905 h
11:20 a.m.	1120 h
Noon	1200 h
4:17 p.m.	1617 h
8:30 p.m.	2030 h
11:35 p.m.	2335 h

Exercise 10.1 Time & Timetables

1. Calculate the number of minutes in
   a) 2 hours; b) 3.6 hours; c) 12 hours; d) 1 day; e) 1 week;
   f) January (31 days); g) February 2016.

2. How many minutes are there between

   a) 7:15 a.m. and 7:35 a.m.? b) 7:15 a.m. and 9:00 a.m.?
   c) 7:15 a.m. and 9:20 a.m.? d) 7:15 a.m. and noon?
   e) 7:15 a.m. and 2:15 p.m.? f) 7:15 a.m. and 6:40 p.m.?

3. Convert to minutes:
   a) 4 hours; b) 4 hours, 37 minutes; c) 10 hours, 9 minutes;

   Also, convert to seconds:
   d) 4 minutes; e) 4 minutes, 37 seconds; f) 10 minutes, 9 seconds.

4. Express the following in hours and minutes.
   For example, 125 minutes = 2 hours, 5 minutes
   a) 65 minutes; b) 130 minutes; c) 100 minutes;
   d) 664 minutes; e) 800 minutes; f) 1000 minutes.

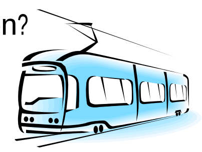
5. Express the following 12-hour times as 24-hour times:
   a) 2.30 a.m. b) 12 noon c) 6.56 a.m.
   d) 12.45 p.m. e) 7.56 p.m. f) 10.50 p.m.

6. Express the following 24-hour times as 12-hour times:
   a) 0430 h b) 1056 h c) 1430 h
   d) 1845 h e) 2020 h f) 2310 h

7. Two friends, P and Q, will meet at the main train station in the city.

   P will walk to her nearest bus stop (5 minutes).
   The bus will take her to the local train station, a 15 minute ride. Then she will take the train into the city (25 minutes).
   a) How long will it take P to travel from home into the city train station?

   Q will cycle 20 minutes from home to her local train station and then take the train into the city (20 minutes).
   b) How long will it take Q to travel from home into the city train station?



   They both leave home at 5:45 p.m.
   c) At what time should each of them reach the city train station?

Exercise 10.1 continued

8. Complete the table below to show times on the 12-hour clock and the equivalent times on the 24-hour clock:

12-hour clock	24-hour clock
12:06 a.m.	0006 h
4:11 a.m.	
9:20 a.m.	
	1015 h
	1156 h
Noon	1200 h
1:40 p.m.	
	1916 h
	2025 h
10:42 p.m.	
11:27 p.m.	
Midnight	0000 h

9. The table shows an early time and a late time. Complete the table to show the time difference (in hours and minutes) between the early time and the late time.

EARLY TIME	LATE TIME	TIME DIFFERENCE
4:15 a.m.	8:18 a.m.	4 hours 3 minutes
4:15 a.m.	10:30 a.m.	
10:25 a.m.	midday	
10:25 a.m.	2:30 p.m.	
11:35 a.m.	5:05 p.m.	
8:20 p.m.	11:30 p.m.	
9:45 p.m.	midnight	
9:45 p.m.	4:22 a.m.	

10. How many minutes are there between the earlier time (mentioned first) and the later time?

a) 6:15 a.m. and 8:05 a.m.

b) 9:15 a.m. and 12:45 p.m.

c) 8:30 p.m. and 1:15 a.m.

d) 0915 h and 1200 h

e) 0915 h and 1425 h

f) 2345 h and 0720 h

11. Give the time on a 12-hour clock

a) 11 minutes after 0520 h;

b) 3 hours, 5 minutes after 0755 h;

c) 4 hours, 40 minutes after 1500 h;

d) 5 hours, 30 minutes after 1220 h;

e) 13 hours after 0417 h;

f) 20 minutes before 1350 h;

g) 25 minutes before 1755 h;

h) 1 hour, 5 minutes before 0840 h;

i) 2 hours, 15 minutes before 0217 h;

j) 3 hours, 20 minutes before 0215 h.

12. During summer, the time in Sydney is 3 hours ahead of Perth time, but in Darwin the time is only 2 hours ahead of Perth time.
Complete this table:

SUMMER TIME

PERTH	DARWIN	SYDNEY
1:00 p.m.		
0230 h	0430 h	
4:20 p.m.		7:20 p.m.
8:15 a.m.		
1730 h		
6:00 a.m.	8:00 a.m.	9:00 a.m.
1200 h		
1930 h	2130 h	

13. In the 2012 London Olympics, USA swimmer, Michael Phelps, just missed out on a gold medal in the 200-metre butterfly. His time was 1 minute, 53.01 seconds.

The medals went to:

Gold:	Chad le Clos (South Africa)	1:52.96
Silver:	Michael Phelps (USA)	1:53.01
Bronze:	Takeshi Matsuda (Japan)	1:53.21

a) What was the margin (time difference) separating le Clos and Phelps?
b) What was the margin separating Phelps and Matsuda?
c) What was the difference between the gold medal and bronze medal times?

For the 200-metre butterfly, each swimmer completed four laps of the pool.

d) For each of the medallists, find the average time per lap.
Give your answers to the nearest hundredth of a second.

Exercise 10.1 continued

14. A clothing workshop employs people to work two shifts per day.

SHIFT	START & FINISH TIMES	TOTAL HOURS
Early	7:00 a.m. to 9:45 p.m.	2¾
Morning	10:00 a.m. to 12:45 p.m.	
Afternoon	1:30 p.m. to	2¾
Evening	6:00 p.m. to	2¾

a) Complete the table.

Between shifts there are breaks for morning tea, lunch and dinner.

b) Complete the table below to show the times and duration of these breaks:

	BREAKS	DURATION
MORNING TEA	9:45 a.m. to 10:00 a.m.	minutes
LUNCH	12:45 p.m. to 1:30 p.m.	45 minutes
DINNER	p.m. to p.m.	minutes

Some workers have to take their children to school and pick them up.

c) What shifts would these workers choose?

15. A summer flight from Perth to Adelaide is scheduled to depart at 1015 h, Perth time.

a) What is the equivalent time on the 12-hour clock?

The flight will take 2 hours 45 minutes.

b) What will be the time in Perth when the plane lands?
Give your answer in both 12-hour time and 24-hour time.

In summer, Adelaide time is 2½ hours ahead of Perth time.

c) What will be the time in Adelaide when the plane departs?

d) What will be the time in Adelaide when the plane lands?

16. A men's downhill Alpine skiing event was held at the Sochi Winter Olympics.
The competitor who completed the course in the shortest time won the gold medal.

Travis Ganong (USA), for example, completed the course in 2:06.64
This notation means 2 minutes and 6.64 seconds

The five fastest downhill skiers in this event were

Aksel Lund Svindal (Norway)	2:06.52
Travis Ganong (USA)	2:06.64
Matthias Mayer (Austria)	2:06.23
Christof Innerhofer (Italy)	2:06.29
Kjetil Jansrud (Norway)	2:06.33

a) Who won the gold medal? the silver medal? the bronze medal?

b) What was the difference between the gold medal and silver medal times?

c) What was the difference between the silver medal and bronze medal times?

d) What time separated the gold medallist from the competitor, Roberts Rode (Latvia) who completed the course in 2:17.50.

17. In February, Donna took flights from Perth to Hobart via Sydney.

Her flight left Perth at 0735 h (local time) and arrived in Sydney at 1450 h (local time).

a) Given that Sydney time was three hours ahead of Perth time, what was the duration of the flight?

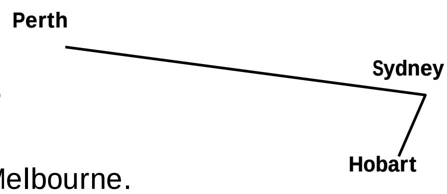
She spent the night in Sydney and left on a flight the next evening at 1940 h to arrive in Hobart at 2135 h.

b) Given that there is no difference between Sydney time and Hobart time, what was the duration of the flight?

Also,

c) how long did she spend in Sydney?

d) how much time altogether did she spend in flight?



On her return, Donna flew from Hobart to Perth via Melbourne.
The flight left Hobart at 1030 h (local time) and arrived in Melbourne 1¼ hours later

e) What was the time in Melbourne?
Note: There is no difference between Melbourne time and Hobart time.

She left Melbourne on a flight which left at 1340 h (Melbourne time) and arrived in Perth 3 hours 55 minutes later.

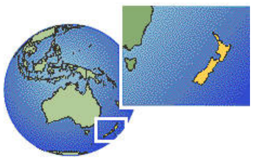
f) What would be the time in Perth when she landed?
Note: Melbourne time was three hours ahead of Perth time.

18. The time in Perth is 3.45 p.m. on a Wednesday afternoon in January.
The table shows the time in various other cities:

Melbourne	6:45 p.m.	Sydney	6:45 p.m.
Adelaide	6:15 p.m.	Darwin	5:15 p.m.
Cape Town	9:45 a.m.	Bangkok	2:45 p.m.
Denpasar	3:45 p.m.	New Delhi	1:15 p.m.
Colombo	1:15 p.m.	Wellington	8:45 p.m.
Beijing	3:45 p.m.	Kuala Lumpur	3:45 p.m.
London	7:45 a.m.	Los Angeles	11.45 p.m. Tuesday
Singapore	3:45 p.m.	Mauritius	11.45 a.m.

a) Which of the cities is in the same time zone as Perth?
What is the time difference between
b) Perth and Adelaide?
c) Perth and Darwin?
d) Perth and Colombo?
e) Perth and Cape Town?
f) Also, list the cities that are three hours ahead of Perth's time?

Wellington is the capital city of New Zealand.



g) Is Wellington's time ahead or behind Perth's time?
By how much?

Sanjeet lives in Perth. He phoned a friend in New Delhi at 10:00 a.m., New Delhi time.

h) What would have been the time in Perth when he made the phone call?

19. Chelsea works at a large shopping centre near the Morley Bus Station.
She catches a #60 bus from the Esplanade Busport (Perth) to get to work by 8.25 a.m.

Route 16, 60 - From Perh — **Monday to Friday**

Timed Stops / Stop No.		12202	12863	16726	16591	11362
Route No.		Esplanade Busport	William St / Forrest St	Hamer Pde / Tenth Av	Waverley St / Pitt St	Morley Bus Stn
Monday to Friday						
am	60	6:29	6:38	6:42	-	6:52
	60	6:57	7:08	7:12	-	7:22
	60	7:23	7:34	7:39	-	7:52
	60	7:38	7:52	7:58	-	8:12
	60	7:58	8:12	8:18	-	8:32
	60	8:13	8:27	8:33	-	8:47
	60	8:35	8:48	8:54	-	9:07
	16	8:37	8:50	8:55	9:15	-
	60	8:51	9:04	9:09	-	9:22
	60	9:07	9:19	9:24	-	9:37
	60	9:22	9:34	9:39	-	9:52
	16	9:31	9:43	9:48	10:08	-
	60	9:37	9:49	9:54	-	10:07

a) What is the latest bus that she should catch from the busport?

Chelsea returns to the busport on an inward bound #60 bus leaving Morley at 4.55 p.m.

b) At what time is she most likely to reach the Esplanade Busport?

20. Kate will travel from Kambalda to Perth next Friday. She will take a bus from Kambalda to Kalgoorlie railway station and then "The Prospector" (a high-speed train) from Kalgoorlie to East Perth terminal.

Location - *(click for details)*	M&W	Fri
Kalgoorlie	14:30	14:30
Coolgardie		15:00
Kambalda	15:14	
Widgiemooltha	15:45	16:00
Norseman	16:45	16:50
Salmon Gums	18:15	18:30
Grass Patch	18:35	18:50
Gibson	19:10	19:25
Esperance	19:30	19:45
	M&W	Fri

Location - *(click for details)*	W&F	Sun
Esperance	08:35	14:00
Gibson	08:55	14:20
Grass Patch	09:30	14:55
Salmon Gums	09:50	15:15
Norseman	10:50	16:45
Widgiemooltha	12:20	17:45
Kambalda	12:47	
Coolgardie		18:45
Kalgoorlie	13:35	19:15
	W&F	Sun

The bus timetable is shown (above).

a) What time on Friday will Kate catch the bus?

The bus will drop Kate at the stop outside Kalgoorlie railway station.

b) At what time will Kate get off the bus? How long will the bus trip take?

The weekly time table for the Prospector (Kalgoorlie to East Perth terminal) is shown:

		Mon Wed Fri	Mon	Tue Thur	Fri	Sat	Sun
From Kalgoorlie		KPL	KPL2	KPA	KPL4	KPA6	KPA4
		AM	PM	AM	PM	AM	PM
Kalgoorlie Station ♿	Dep	7:05	3:00	7:05	3:00	7:05	2:05
Bonnie Vale	Dep	7:23	3:18	7:23	3:18	7:23	2:23
Southern Cross	Dep	9:12	5:07	9:12	5:15	9:17	4:12
Moorine Rock	Dep	9:29	5:24	9:29	5:36	9:34	4:29
Bodallin	Dep	9:41	5:36	9:41	5:48	9:46	4:41
Carrabin	Dep	9:54	5:49	9:54	6:01	9:59	4:54
Burracoppin	Dep	10:06	6:01	10:06	6:12	10:11	5:06
Merredin	Dep	10:23	6:18	10:23	6:29	10:28	5:23
Doodlakine	Dep	NS	6:48	10:53	6:59	10:58	5:53
Kellerberrin	Dep	NS	6:57	11:02	7:08	11:07	6:02
Tammin	Dep	NS	7:15	11:20	7:26	11:25	6:20
Cunderdin	Dep	NS	7:28	11:35	7:39	11:38	6:33
Meckering	Dep	NS	7:42	11:49	7:53	11:52	6:47
Northam	Dep	12:09	8:02	12:09	8:13	12:12	7:07
Toodyay	Dep	NS	NS	12:29	NS	12:29	7:28
Midland	Dep	1:23	9:15	1:23	9:25	1:23	8:20
East Perth Terminal ♿	Arr	1:45	9:35	1:45	9:45	1:45	8:40

c) At what time will Kate catch the train?

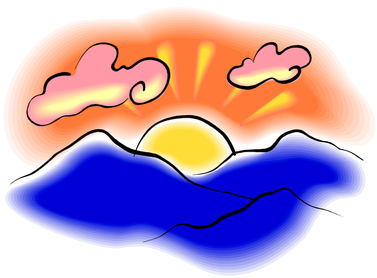
d) When will she get to East Perth terminal?

e) Approximately how long did it take Kate to get from Kambalda to Perth?

Exercise 10.1 continued

21. The table shows the times of sunrise and sunset in Perth for each day in March 2014.

The time difference between sunrise and sunset is also given.



Date	Sunrise	Sunset	Difference
1 Mar 2014	6:06 AM	6:52 PM	12h 46m 10s
2 Mar 2014	6:06 AM	6:51 PM	12h 44m 12s
3 Mar 2014	6:07 AM	6:49 PM	12h 42m 15s
4 Mar 2014	6:08 AM	6:48 PM	12h 40m 17s
5 Mar 2014	6:09 AM	6:47 PM	12h 38m 20s
6 Mar 2014	6:09 AM	6:46 PM	12h 36m 22s
7 Mar 2014	6:10 AM	6:45 PM	12h 34m 23s
8 Mar 2014	6:11 AM	6:43 PM	12h 32m 25s
9 Mar 2014	6:12 AM	6:42 PM	12h 30m 27s
10 Mar 2014	6:12 AM	6:41 PM	12h 28m 28s
11 Mar 2014	6:13 AM	6:40 PM	12h 26m 30s
12 Mar 2014	6:14 AM	6:38 PM	12h 24m 31s
13 Mar 2014	6:15 AM	6:37 PM	12h 22m 32s
14 Mar 2014	6:15 AM	6:36 PM	12h 20m 34s
15 Mar 2014	6:16 AM	6:35 PM	12h 18m 35s
16 Mar 2014	6:17 AM	6:33 PM	12h 16m 36s
17 Mar 2014	6:17 AM	6:32 PM	12h 14m 38s
18 Mar 2014	6:18 AM	6:31 PM	12h 12m 39s
19 Mar 2014	6:19 AM	6:30 PM	12h 10m 41s
20 Mar 2014	6:20 AM	6:28 PM	12h 08m 42s
21 Mar 2014	6:20 AM	6:27 PM	12h 06m 44s
22 Mar 2014	6:21 AM	6:26 PM	12h 04m 46s
23 Mar 2014	6:22 AM	6:24 PM	12h 02m 48s
24 Mar 2014	6:22 AM	6:23 PM	12h 00m 50s
25 Mar 2014	6:23 AM	6:22 PM	11h 58m 52s
26 Mar 2014	6:24 AM	6:21 PM	11h 56m 55s
27 Mar 2014	6:24 AM	6:19 PM	11h 54m 57s
28 Mar 2014	6:25 AM	6:18 PM	11h 53m 00s
29 Mar 2014	6:26 AM	6:17 PM	11h 51m 03s
30 Mar 2014	6:26 AM	6:15 PM	11h 49m 06s
31 Mar 2014	6:27 AM	6:14 PM	11h 47m 10s

On which date was

a) the earliest sunrise in March 2014?

b) the latest sunrise in March 2014?

On how many days in March 2014, was there

c) less than 12 hours between sunrise and sunset?

d) more than 12½ hours between sunrise and sunset?

The period between sunset on one day and sunrise the next day is called “night”.

How long (in hours and minutes) was the night of

e) 13 March 2014?

f) 30 March 2014?

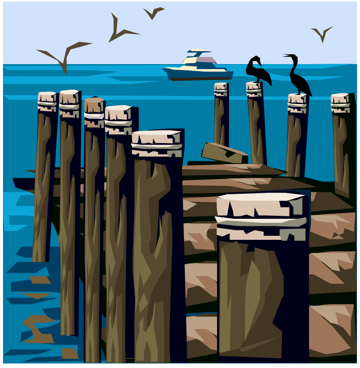
An equinox is a day in the year when day and night are of equal length: that is, 12 hours each or very close to it.

g) Was there an equinox in March 2014? Which day?

There are two equinoxes in every year.

h) During which month in 2014 was the other equinox?

22. The tide chart for Esperance WA gives the time and height of low and high waters for the first week in March 2014:



TIDES (Esperance WA)

	Low	High	Low	High
Sat Mar 1:	0.23 m at 0546 h	0.76 m at 1155 h	0.33 m at 1710 h	0.98 m at 2329 h
Sun Mar 2:	0.28 m at 0602 h	0.79 m at 1205 h	0.34 m at 1749 h	0.88 m at 2341 h
Mon Mar 3:	0.34 m at 0617 h	0.83 m at 1207 h	0.38 m at 1826 h	0.77 m at 2353 h
Tue Mar 4:	0.38 m at 0617 h	0.87 m at 1222 h	0.45 m at 1859 h	0.68 m at 2352 h
Wed Mar 5:	0.37 m at 0532 h	0.89 m at 1241 h	0.52 m at 1922 h	0.66 m at 2258 h
Thu Mar 6:	0.34 m at 0531 h	0.89 m at 1249 h	0.58 m at 1918 h	0.66 m at 2305 h
Fri Mar 7:	0.30 m at 0517 h	0.89 m at 1249 h	0.64 m at 1940 h	0.66 m at 2235 h

Use the tide chart to answer the following questions:

a) How many high tides were there each day? How many low tides?

b) At what time(s) was there high tide on Thursday March 6th?

c) If you wished to depart Esperance by boat on a high tide on Tuesday March 4th, what's the earliest time you could leave?

d) You were ready to leave Esperance by boat at 1:30 p.m. on Friday 7th March but had to wait for high tide. What's the earliest that you could leave?

e) What was the time interval between the two high tides on Thursday March 6th?

f) "There's always at least 12 hours between low tides!"
List the evidence from the tide chart to show that this statement is not true.

g) For the second week in March 2014, when, in general terms, could high tides be expected? When could low tides be expected?